Lead R5.4: Conjecture 10 over prime fields, and the oracle-dependence of the saturation law

2026-10-12. Theory round with native sanity checks (`examples/r5_4_theory_checks.rs`). Full proofs in `RESULTS.md`; toy instances only; no sub-rho claim; no claim about any deployed curve.

Verdict

(i) Conjecture 10. Over $\mathbb{F}_p$ every known attack family (generic/rho with automorphisms, MOV/Frey–Rück, anomalous, Weil descent/GHS, summation-polynomial index calculus on curve-independent x-sets) has a cost equal to a function of (p, t) times a member factor bounded by a constant: $\sqrt{3}$ for rho (unit groups of size ≤ 6), 1 for pairings and anomalous attacks, not applicable for descent, $\leq 6 \cdot h$ for index calculus (heuristic). **Theorem 4:** if two members' endomorphism rings differ only at primes $\leq \Delta$, DLP transfers between them at per-instance cost $\mathrm{poly}(\Delta, \log p)$ (vertical part unconditional, horizontal under GRH). So Conjecture 10 can fail only across a prime conductor gap $\ell \geq p^{\varepsilon}$. **Measured:** such classes are a fraction $1.7 \cdot 10^{-2}$ ($\varepsilon = 0.1$), $2.7 \cdot 10^{-4}$ (0.25), $3.8 \cdot 10^{-6}$ (0.4) of all 1048576 ordinary traces at $p \approx 2^{36}$. Open (Conjecture 10′): is there a prime-field index calculus whose relation probability depends on the conductor? Over $\mathbb{F}_{2^N}$, N composite, and $\mathbb{F}_{p^k}$, $k > 1$, the failing hypothesis is the existence of proper subfields (GHS genus depends on the member's coefficients).

(ii) Saturation law. For an oracle sweeping $(m-k)$-tuples from a domain Ω with a k-summand subsolver, a kernel-saturated base pays $\mu^{m-2} \cdot \left(|\Omega| / |F|\right)^{m-k}$ (Prop. 15.2′). Sweeping F: μ^{m-2}. Sweeping a transversal of F/M: μ^{k-2}, i.e. **1 for the pair-table mitm oracle**. The domain ratio μ^{m-k} is exact (Lemma 7). A1-R's $2.95 \times = 2.17$ (redundant sweep, law 2) $\times$ 1.36 (index-2 image base, law 1.51); the law with $\Omega = F$ predicts 3.02. **Characteristic 2 (Theorem 9):** the quotient by $T = \left(0, \sqrt{b}\right)$ is the Frobenius conjugate, $x(2P) = u(P)^2$ with $u = x + \sqrt{b}/x$, so $[2](F_u) = F_{\mathrm{tz}}$: the quotient-domain u oracle **is** the trace-zero oracle, and any algebraic system in a T-invariant coordinate is the plain x-frame system with target $x(2R)$. 0 exceptions on 30 bases over 10 binary curves.

Figures

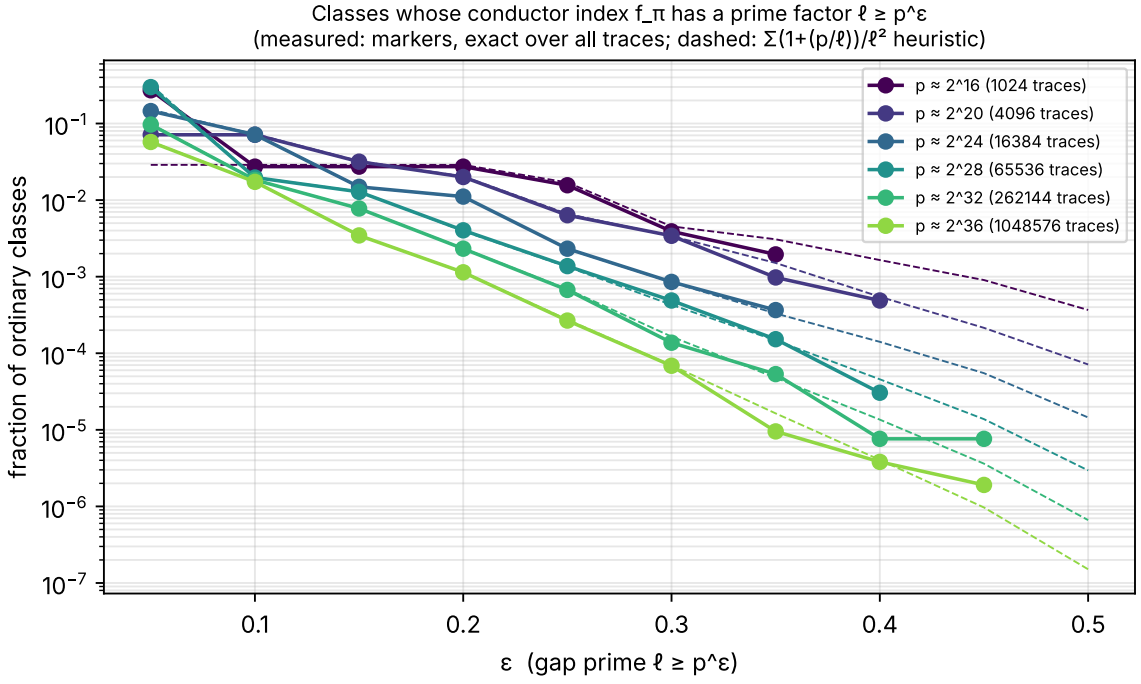


Figure 1: Fraction of ordinary isogeny classes over $\mathbb{F}_p$ (all traces, one prime per size) whose conductor index has a prime factor $\ell \geq p^{\varepsilon}$: the only region where Conjecture 10 is not reduced to Theorem 4. Measured (markers, exact) and the union-bound heuristic (dashed). Data: `conductor.json`.

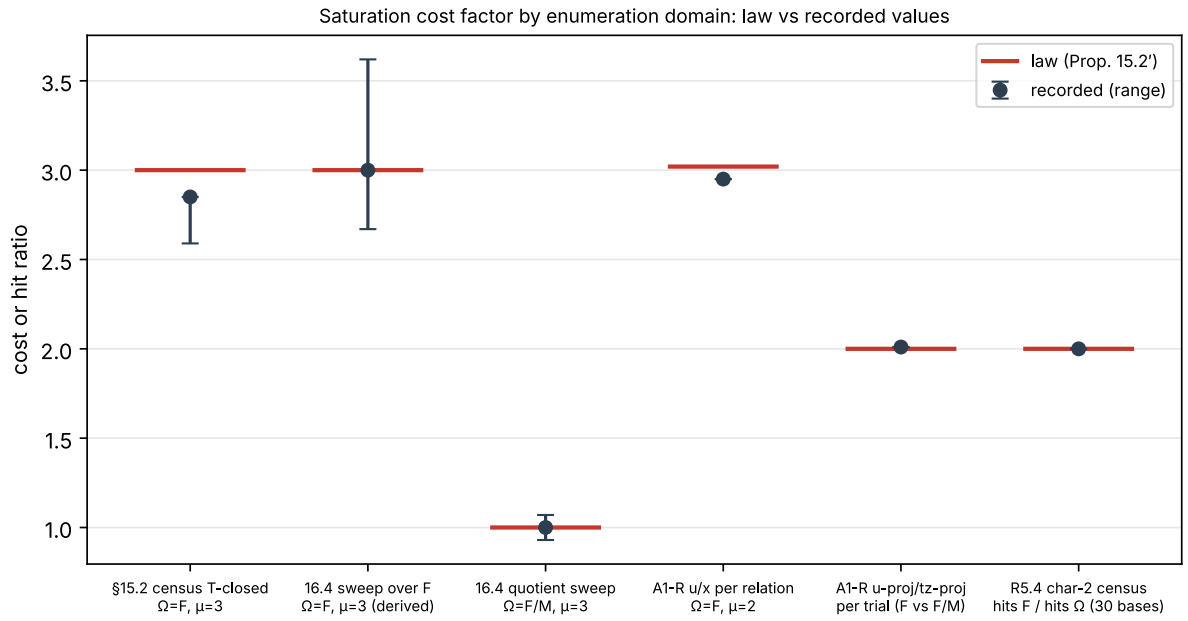


Figure 2: Saturation cost and hit ratios: Prop. 15.2′ (red) against recorded values from round 3 (§15.2 census), round 4 (16.4), §7.A1-R and this round's exact char-2 census (`char2.json`). Ranges are the recorded spreads across curves.

Exact checks (measured, `char2.json`)

check	instances	exceptions
$x(2P) = u(P)^2$; halving criterion $P \in 2E \Leftrightarrow \mathrm{Tr}(x) = \mathrm{Tr}(a)$	every point, 10 curves, $n \in \{11, 13, 17\}$	0
$[2](F_u) = F_{\mathrm{tz}(V^2)}$; D_m^G equal ($m = 2, 3$); columns equal	30 bases	0
transversal sweep finds every decomposable target; $\mathrm{hits}_F = 2 \cdot \mathrm{hits}_\Omega$	30 bases, all of G	0

Predictions for R5.1 (A1 cell, $m = 3$)

P1 exact: $[2](F_u)$ equals the tz base, 19 = 19 columns. P2 exact: a transversal sweep halves hits and lookups, same decomposable targets. P3: the quotient-domain u arm costs about the tz arm, $\approx 4.8 \cdot 10^3$ GAE per independent relation (vs 10 387), S ≈ 85 (vs 140). P4: $m = 2$ unchanged. P5: a u-symmetrised F4 oracle does the x-frame oracle's work per call.

What this does not say

Not a proof of Conjecture 10 (no non-generic lower bounds exist); Theorem 4′s horizontal part is conditional on GRH / heuristic path finding; cost factors other than the exact domain ratio are heuristic; Corollary 9.2 covers Weierstrass binary curves and the 2-torsion point only, not binary Edwards or 4-torsion formulations. Literature marked "recalled" in `RESULTS.md` was not re-opened.